

|      |                                                                                                                                                                                                                                                                                                                                                                             |       |       |       |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------|
|      | <b>DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE</b><br><b>Supplementary Summer 2024</b><br><b>Course: B. Tech. Branch : AI &amp; DS and Allied Branches Semester :III</b><br><b>Subject Code &amp; Name:BTESC304 Computer Architecture and Max Marks: 60</b><br><b>Operating System</b><br><b>Date: 06/07/2024 Duration: 3 Hrs.</b>                              |       |       |       |
|      | Instructions to the Students:<br>1. All the questions are compulsory.<br>2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.<br>3. Use of non-programmable scientific calculators is allowed.<br>4. Assume suitable data wherever necessary and mention it clearly. |       |       |       |
|      |                                                                                                                                                                                                                                                                                                                                                                             | CO    | Level | Marks |
| Q. 1 | Solve Any Two of the following                                                                                                                                                                                                                                                                                                                                              |       |       | 12    |
| A)   | Difference between computer architecture and computer organization.                                                                                                                                                                                                                                                                                                         | (CO1) | BL4   | 6     |
| B)   | Explain different types of addressing modes.                                                                                                                                                                                                                                                                                                                                | (CO1) | BL2   | 6     |
| C)   | Explain functional unit of computer system.                                                                                                                                                                                                                                                                                                                                 | (CO1) | BL2   | 6     |
|      |                                                                                                                                                                                                                                                                                                                                                                             |       |       |       |
| Q. 2 | Solve Any Two of the following                                                                                                                                                                                                                                                                                                                                              |       |       | 12    |
| A)   | Explain characteristics of Magnetic Disc, Optimal Memory and RAID.                                                                                                                                                                                                                                                                                                          | (CO2) | BL2   | 6     |
| B)   | What is page replacement algorithm? Explain least recently algorithm.                                                                                                                                                                                                                                                                                                       | (CO2) | BL2   | 6     |
| C)   | Difference between SRAM and DRAM.                                                                                                                                                                                                                                                                                                                                           | (CO2) | BL4   | 6     |
|      |                                                                                                                                                                                                                                                                                                                                                                             |       |       |       |
| Q. 3 | Solve Any Two of the following                                                                                                                                                                                                                                                                                                                                              |       |       | 12    |
| A)   | Difference between Hardwired control unit and Micro-programmed control unit.                                                                                                                                                                                                                                                                                                | (CO3) | BL4   | 6     |
| B)   | Explain instruction pipelining.                                                                                                                                                                                                                                                                                                                                             | (CO3) | BL2   | 6     |
| C)   | Write a short note on DMA.                                                                                                                                                                                                                                                                                                                                                  | (CO3) | BL2   | 6     |
|      |                                                                                                                                                                                                                                                                                                                                                                             |       |       |       |
| Q. 4 | Solve Any Two of the following                                                                                                                                                                                                                                                                                                                                              |       |       | 12    |
| A)   | Define an operating system with the help of different component of operating system.                                                                                                                                                                                                                                                                                        | (CO4) | BL2   | 6     |
| B)   | Explain the concept of virtual machines                                                                                                                                                                                                                                                                                                                                     | (CO4) | BL2   | 6     |

|      |                                                                               |       |     |    |
|------|-------------------------------------------------------------------------------|-------|-----|----|
| C)   | Explain process scheduling algorithms.                                        | (CO4) | BL2 | 6  |
|      |                                                                               |       |     |    |
| Q. 5 | Solve Any Two of the following                                                |       |     | 12 |
| A)   | Explain critical section problem in briefly.                                  | (CO5) | BL2 | 6  |
| B)   | What is semaphore? Explain types, advantages and disadvantages of semaphores. | (CO5) | BL2 | 6  |
| C)   | Write a short note on deadlock prevention.                                    | (CO5) | BL2 | 6  |
|      | ***End***                                                                     |       |     |    |